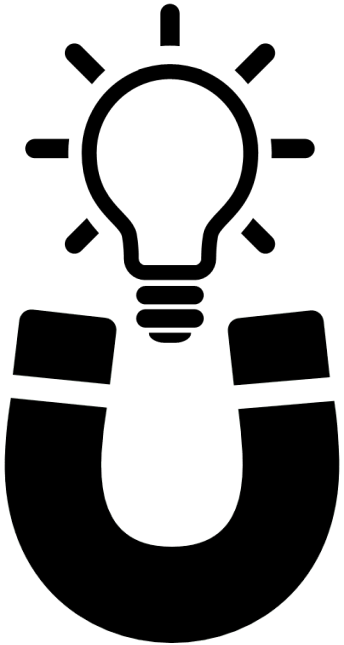
Magnetically Controllable Fluorescent Proteins

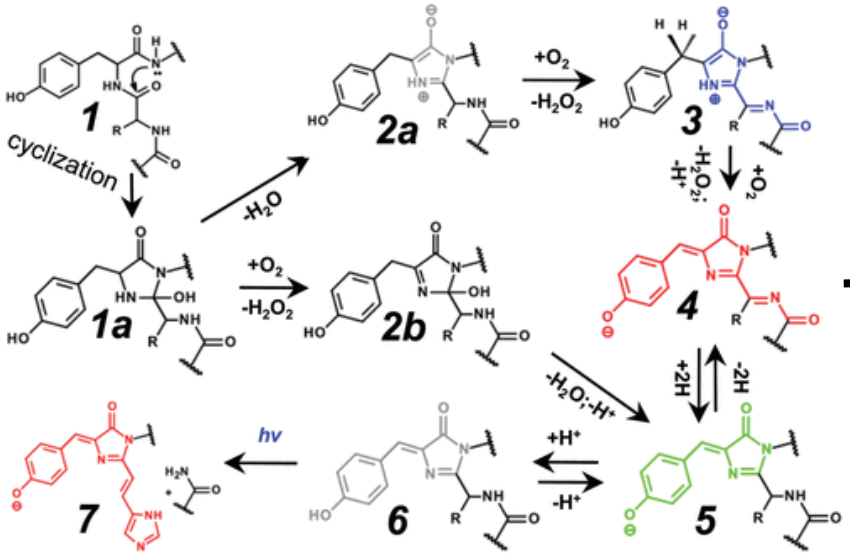


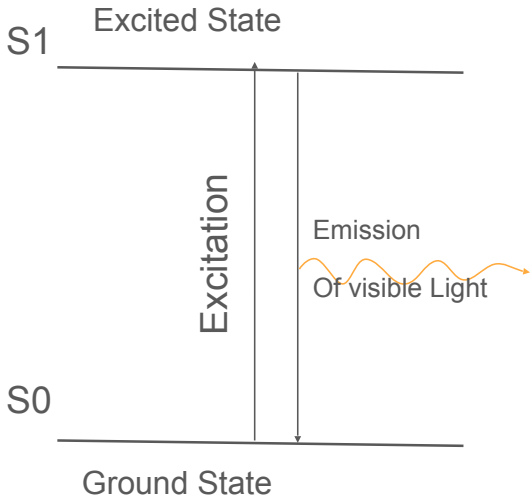
Stanford MEDICINE

SLAC NATIONAL ACCELERATOR LABORATORY

Presentor: Jerome Wu
Mentors: Samsuzzoha Mondal and Soichi Wakatsuki

Excitable Motif

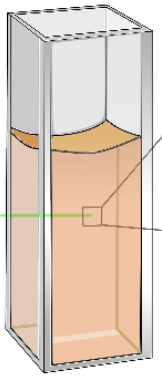


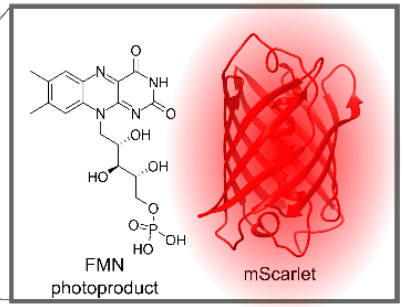


Fluorescent protein

Initial Observation: mScarlet in a mixture with FMN shows MFE (Magnetic Field Effect)

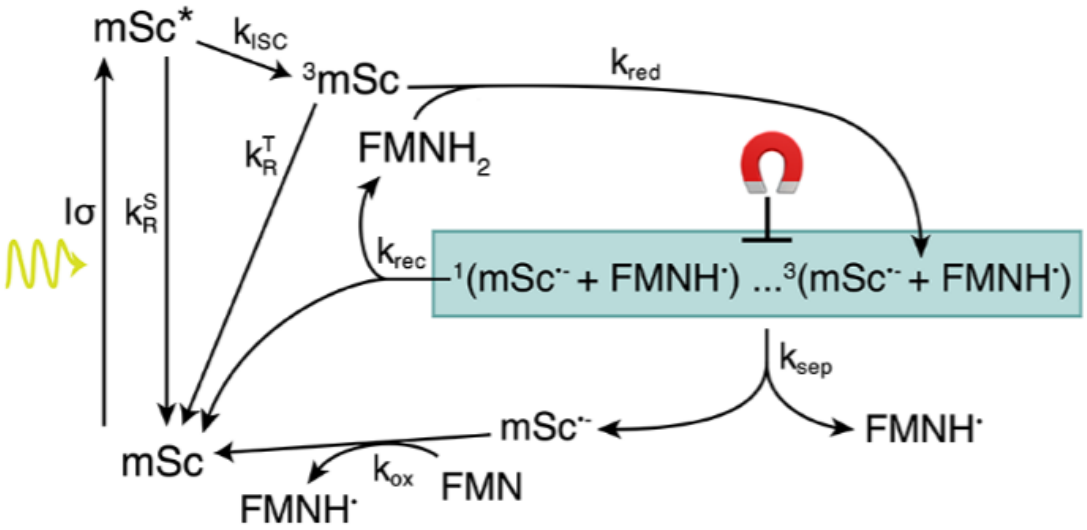
Fluorescent Cofactor





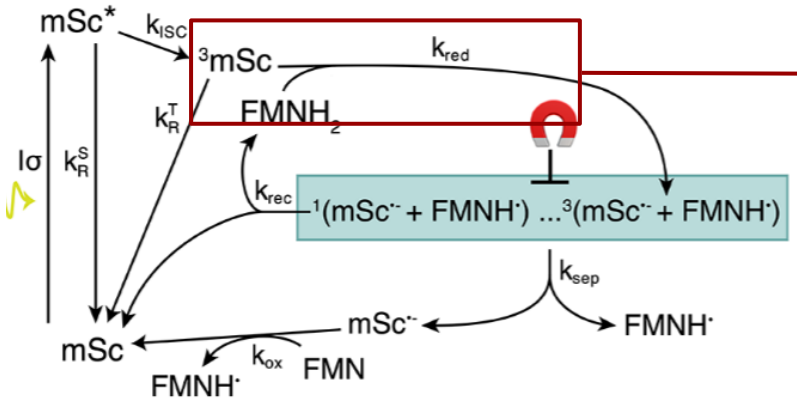
Why Though?

Here's one piece of **Recent** and **Interesting** Conjecture



Xiang, K. M.; Lampson, H.; Hayward, R. F.; York, A. G.; Ingaramo, M.; Cohen, A. E. Mechanism of Giant Magnetic Field Effect in a Red Fluorescent Protein. *J. Am. Chem. Soc.* **2025**, *147* (21), 18088–18099. https://doi.org/10.1021/jacs.5c03997.

Breaking that down a bit 1/6

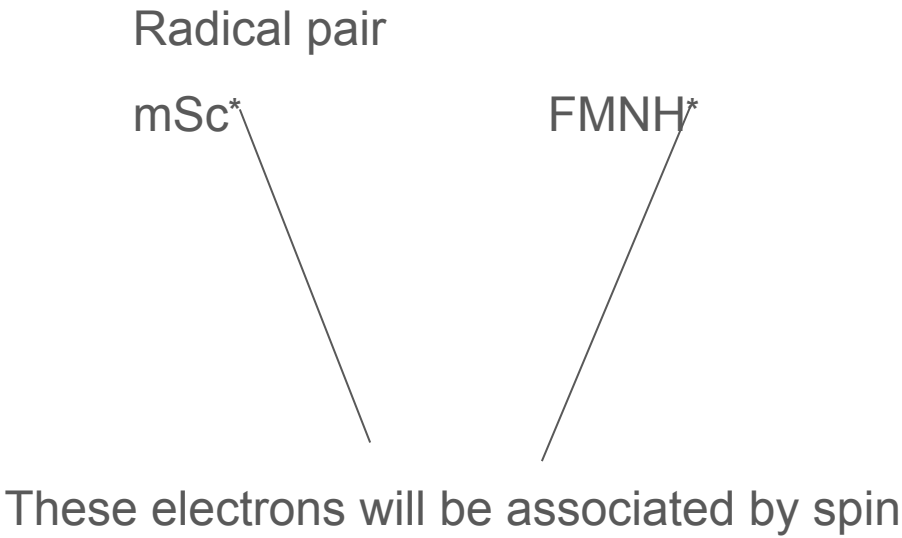


^{3}mSc + FMNH$_2$ = mSc* + FMNH*

e^- ,
H^+

Spin
Correlated
Radical Pair

Breaking that down a bit 2/6



Breaking that down a bit 3/6

Singlet Triplet

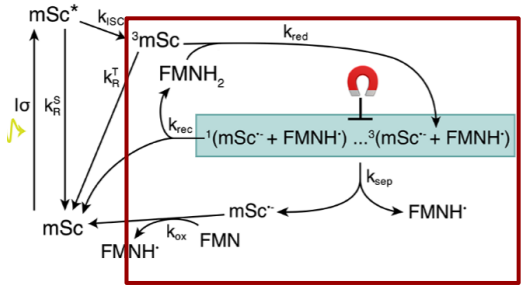
Breaking that down a bit 4/6

Spin Correlated Radical Pair

$mSc^* + FMNH^*$

Favored by singlet state

Favored by triplet state



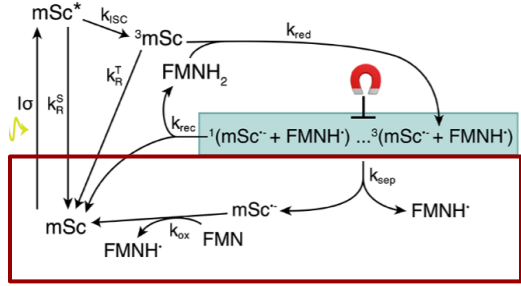
Recombine into:
$mSc + FMNH_2$

Disassociate into mSc^* and $FMNH^*$

Breaking that down a bit 5/6

Spin Correlated Radical Pair

$mSc\cdot + FMNH\cdot$



ed by singlet state

Favored by triplet state

Recombine in

$^3mSc + FMNH_2$

Disassociate into mSc* and FMNH*

$mSc\cdot + FMN = mSc$

Breaking that down a bit 6/6

$mSc^* + FMN = mSc + FMNH\cdot$

Now back to fluorescence…

Much higher population of ground state mSc

Entire radical pair pathway does not fluoresce at all

Magnet on = Less Fluorescence!

How do we start trying to prove this?

If this mechanism really works off of a spin correlated radical pair, then we should be able to detect RYDMR

Burd, S. C.; Bagheri, N.; Ingaramo, M.; Condon, A. F.; Mondal, S.; Dowlatshahi, D. P.; Summers, J. A.; Mukherjee, S.; York, A. G.; Wakatsuki, S.; Boxer, S. G.; Kasevich, M. Magnetic Resonance Control of Reaction Yields through Genetically-Encoded Protein:Flavin Spin-Correlated Radicals in a Live Animal. *bioRxiv* **2025**, 2025.02.27.640669. https://doi.org/10.1101/2025.02.27.640669.

RYDMR

Reaction-Yield Detected Magnetic Resonance

Why though?

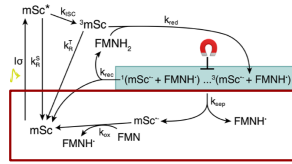
Spin Correlated Radical Pair

Also have singlet and triplet states

mSc· + FMNH·

Favored by singlet state

Favored by triplet state

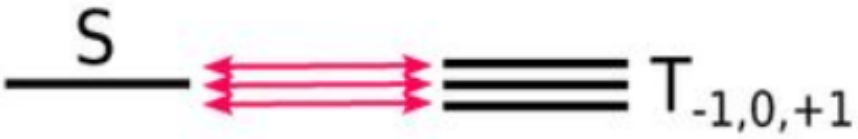


Recombine in
^{3}mSc + FMNH

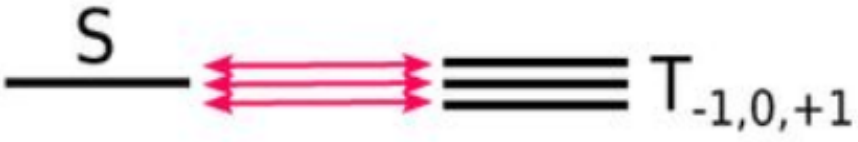
Disassociate into mSc· and FMNH·

mSc· + FMN = mSc

Normally:

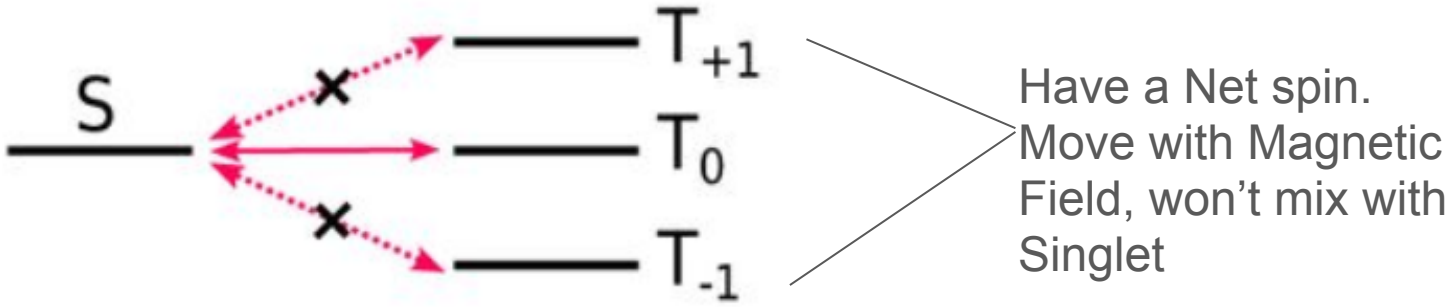


Normally:



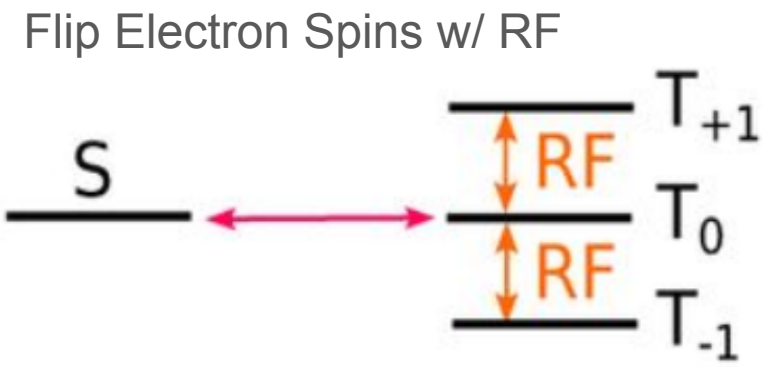
Normal Fluorescence!

Under 50-60mT



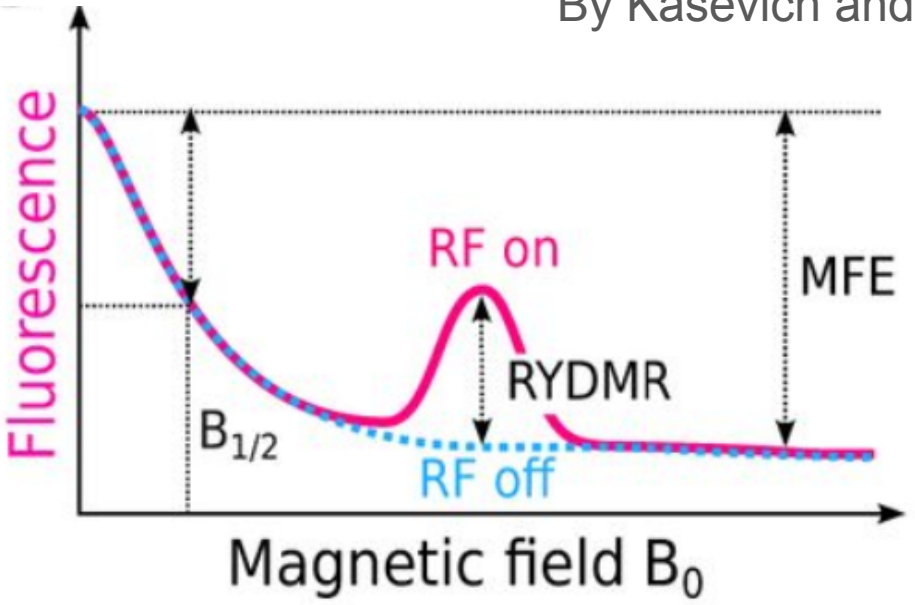
Application of a RF (Radio Frequency) Field at ESR (Electron Spin Resonance) Frequency

Application of a RF Field at ESR Frequency



This has been Observed!

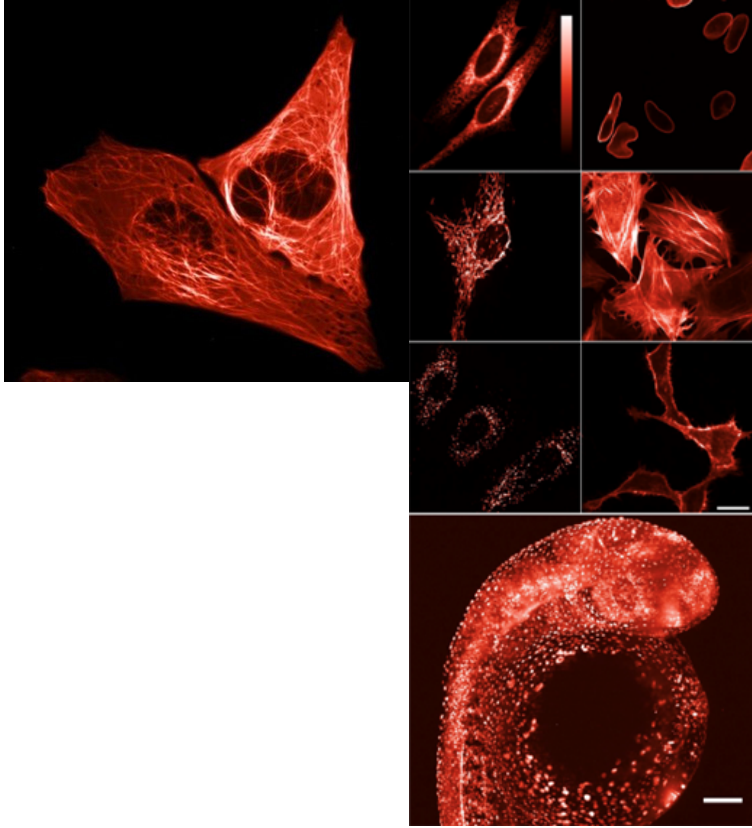
By Kasevich and Boxer and Us

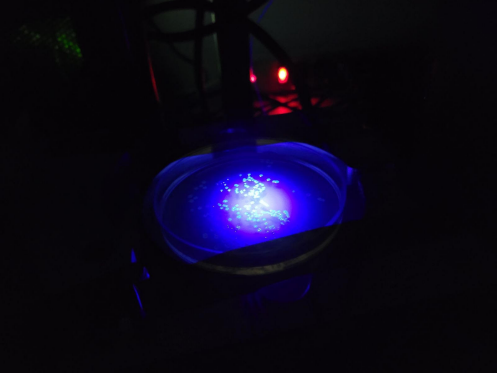


Applications

Imaging w/ Controllable brightness?

Turn a gene on or off with a magnet?





It’s a Good Time to be Working on MFE!